



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE(Advanced)
PRE-SCORE
05-11-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS, TAAS & TTAS)

Time : 3 Hours

PAPER-2 (OPTIONAL)

Maximum Marks : 180

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their Form No. in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

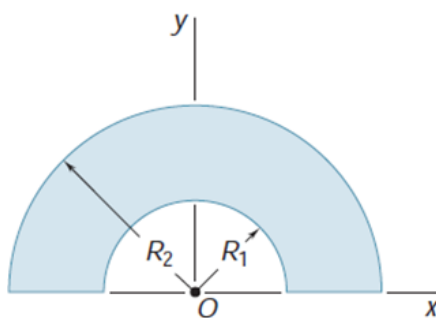
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. A semi-annular disc of mass m has dimensions as shown. Find the moment of inertia of the disc about different axes.



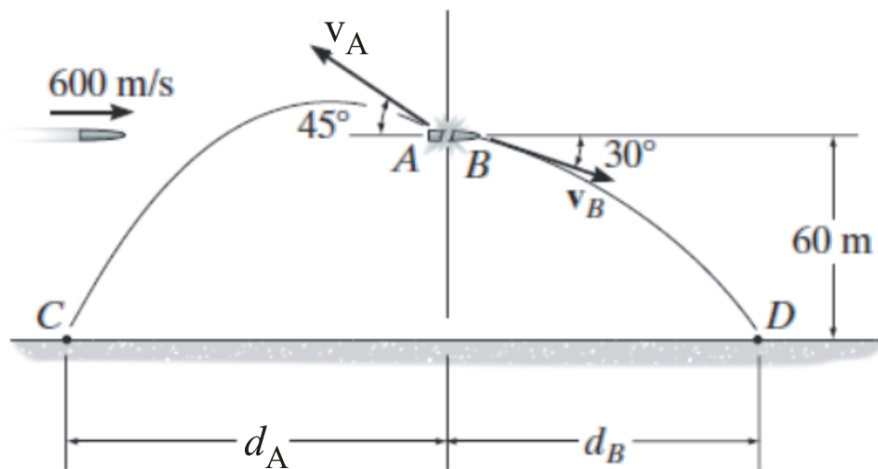
(A) $I_x = \frac{M (R_1^2 + R_2^2)}{4}$

(B) $I_y = \frac{M (R_1^2 + R_2^2)}{4}$

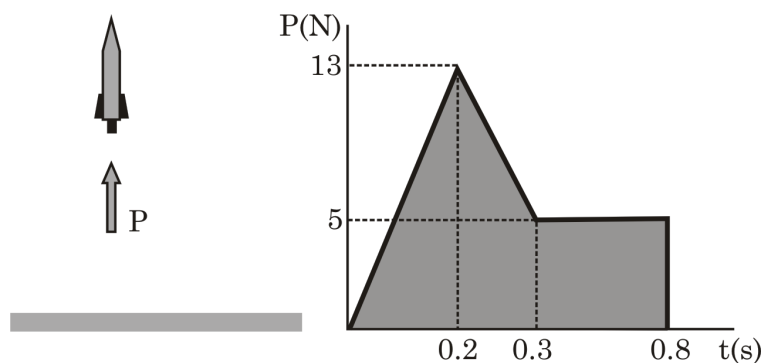
(C) $I_z = \frac{M (R_1^2 + R_2^2)}{2}$

(D) I about any axis in the x-y plane passing through the origin = $\frac{M (R_1^2 + R_2^2)}{4}$

2. A 4 kg projectile travels with a horizontal velocity of 600 m/s before it explodes and breaks into two fragments A and B of mass 1.5 kg and 2.5 kg, respectively. If the fragments travel along the parabolic trajectories shown,

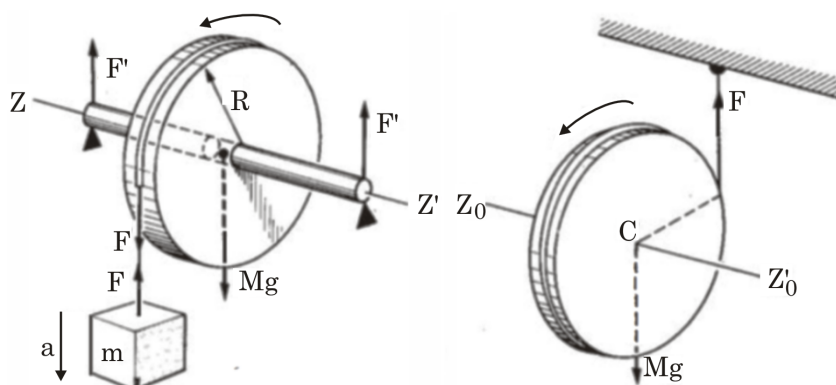


- (A) Both the fragments hit the ground simultaneously
 (B) A reaches ground after B
 (C) $V_A = 800 (\sqrt{6} + \sqrt{2})$ m/s
 (D) $V_B = 960 (\sqrt{3} + 1)$ m/s
3. A 100 gm rocket is fired vertically. The engine applies a thrust P which varies in magnitude as shown. Neglecting air resistance and the change in mass of the rocket, (Take : $g = 10 \text{ m/s}^2$)

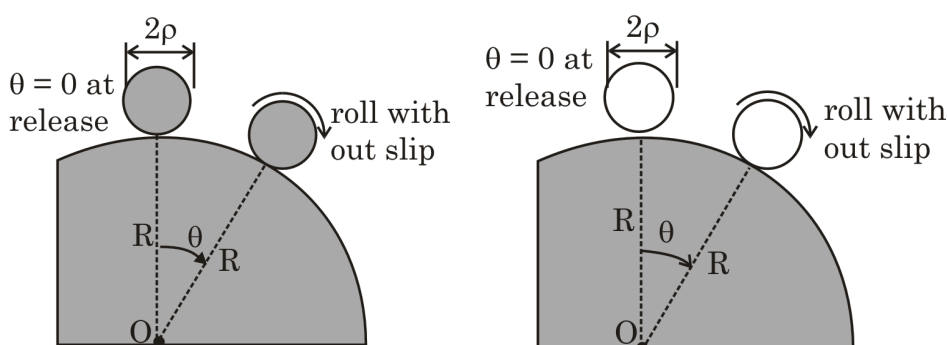


- (A) the maximum speed of the rocket as it goes up is 39 m/s.
 (B) the time for the rocket to reach its maximum elevation is 4.7 sec.
 (C) The total impulse of the engine is 4.7 Ns.
 (D) the impulse of gravity during the time engine acts is 0.8 Ns.

4. Figure 1 shows the block of mass m connected to a disc of mass M as shown. The disc can rotate freely about its massless axle and the string does not slip on the disc. Figure 2 shows the same disc hanging from an inextensible thread. The string does not slip on the disc. The systems are released from rest simultaneously. The tension in both the strings is F as shown.

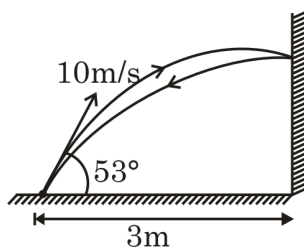


- (A) $m = M$
- (B) In time t , kinetic energy of disc in both the cases is the same
- (C) In time t , the rotational kinetic energy of both the discs is same
- (D) The angular acceleration of disc is same in both the cases.
5. We have a fixed hemispherical surface of radius R . We place a sphere and a ring on the hemisphere one by one and release them. Both of them roll on the surface till they leave the surface.



- (A) The Sphere covers a larger angle θ on the hemisphere.
- (B) The ring has higher kinetic energy at the instant of leaving the hemisphere.
- (C) The sphere has higher speed at the instant of leaving the hemisphere.
- (D) The work done by contact force is same in both the cases.

6. A ball is projected towards a smooth vertical wall 3m away. It collides the wall & lands back at the point of projection. Choose the correct statement(s) :-



- (A) time taken to reach wall is more than time taken to reach back after collision.
 (B) time taken to reach wall is less than time taken to reach back after collision.
 (C) the coefficient of restitution is $\frac{5}{11}$.
 (D) the coefficient of restitution is $\frac{1}{3}$

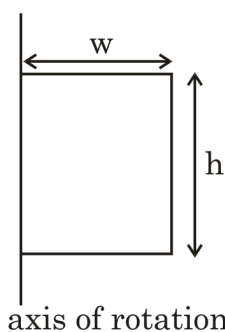
SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

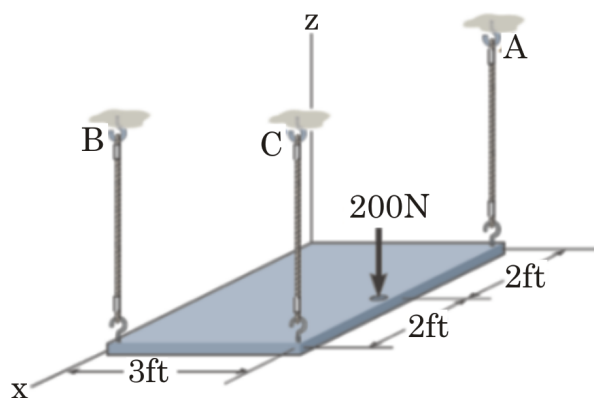
Children try to push the door at any point to open it. But the adult can easily open the door by pushing the door as far from the hinge as possible. That is why we see the handle of the door at the far end, away from the hinge. While the idea can be easily explained by concept of torque and the lever arm, a question arises that if the same force is applied at a point close to the hinge, acceleration of center of mass of door is lesser. Which force reduces this effect ? We will try to answer this question . Consider a uniform rectangular door of height h and width w , mass m . Hinges are smooth. Neglect the effect of gravity.



7. A force F is applied perpendicular to the door at a distance of $\frac{w}{3}$ from hinge. What is the force exerted by the hinge at this moment ?
- (A) $0.5 F$ in direction opposite to the force.
 (B) $0.5 F$ in direction of the force.
 (C) $\frac{F}{3}$ in direction opposite to the force
 (D) $\frac{2F}{3}$ in direction opposite to the force.
8. How far from the axis should the force be applied so that force exerted by hinge is zero at that moment.
- (A) $\frac{w}{2}$ (B) $\frac{3w}{4}$ (C) $\frac{2w}{3}$ (D) $\frac{7w}{12}$

Paragraph for Questions 9 and 10

The uniform plate has a weight of 500 N. Determine the tension in each of the supporting cables. There are only 3 cables supporting the rectangular plate. A force of 200 N is also acting at the mid-point of an edge as shown in figure.



9. Tension in cable C is
 - (A) 100 N
 - (B) 200 N
 - (C) 400 N
 - (D) 500 N
10. Tension in cable B is
 - (A) same as that of A but less than that in C
 - (B) more than that of A but less than that in C
 - (C) less than that of A but more than C
 - (D) same as that of A and more than that in C

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

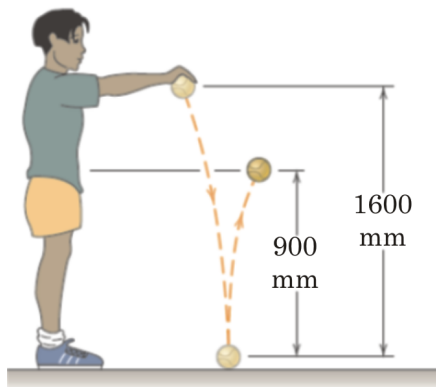
Paragraph for Questions 1 and 2

Two particles of mass 1 kg and 4 kg are moving in empty space with velocity of 1 m/s in negative x direction and 4 m/s in positive x direction respectively. 1 kg mass is at the origin of coordinates at the given instant and 4 kg mass is at $x = 4$ m at that instant. They exert an attractive force on each other which depends on the distance s between them as $F = 5s$.

1. What is the maximum distance $s_{\max}$ (in m) between them during the course of their motion. Fill $(2s_{\max})^2$?
2. What is the distance s (in m) between them when 1 kg mass stops for the first time. Fill $(2s)^2$

Paragraph for Questions 3 and 4

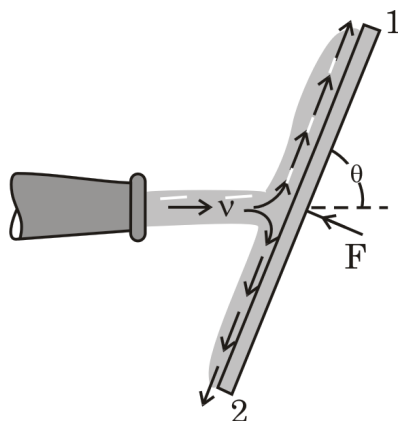
Tennis balls are usually rejected if they fail to rebound to waist level when dropped from shoulder level. A ball just passes the test as indicated in the figure. Assume that the coefficient of restitution between the ball and the floor is the same as the coefficient of restitution between the balls.



3. What is the coefficient of restitution for the collision between the floor and the ball.
4. If two identical tennis balls moving at the same speed in opposite direction collide head on, what is the percentage of total mechanical energy that is lost?

Paragraph for Questions 5 and 6

A jet of fluid with cross-sectional area A and mass density ρ issues from the nozzle with a velocity v and impinges on the inclined trough shown in section. Some of the fluid is diverted in each of the two directions. If the trough is smooth, the velocity of both diverted streams remains v , and the only force which can be exerted on the trough is normal to the bottom surface. Hence, the trough will be held in position by forces whose resultant is F normal to the trough. Take $\theta = 60^\circ$, $\rho = 10^3 \text{ kg/m}^3$, $v = 20 \text{ m/s}$, $A = \sqrt{3} \text{ cm}^2$.



5. Determine the force F (in N) required to support the trough.
6. Also find the volume rates of flow Q_1 and Q_2 for the two streams. What is the ratio Q_1/Q_2 ?

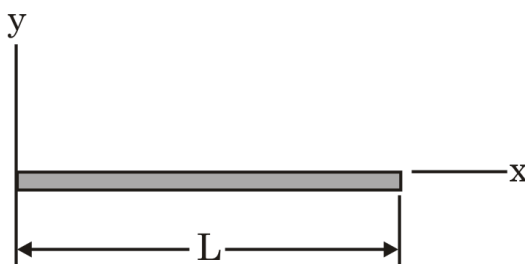
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

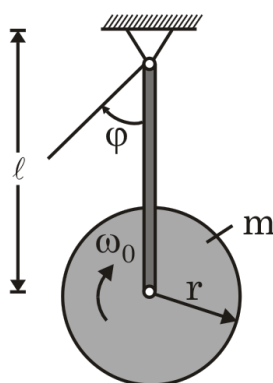
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

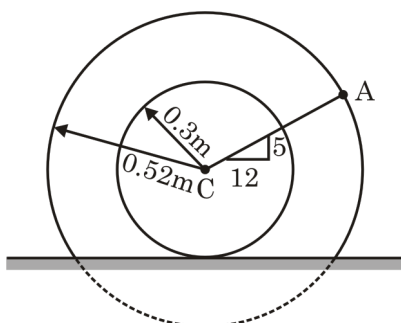
7. A thin rod has variable linear mass density given by $\lambda = \lambda_0 \left(1 + \frac{x^2}{L^2}\right)$. Find the distance of its center of mass from the origin (in cm). Take $L = 8$ m.



8. A homogeneous circular disk of weight $W = mg$ is suspended from a pin-supported bar (of negligible mass). Initially the disk rotates with the angular velocity ω_0 about its centre. There is a catch in the bar which engages the disc and the disc comes to rest with respect to the bar. What should be ω_0 (in rad/s) so that the system may complete the vertical circle. (Take $\ell = 40$ cm, $r = 20$ cm, $m = 1$ kg, $g = 9.8$ m/s²)



9. The wheel shown in Figure rolls with slipping in such a manner that its angular velocity is 5 rad/s in the clockwise direction while the velocity of the centre C is 2 m/s to the left. Determine the speed of point A (in cm/s).



PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
1. A substance which causes pollution is known as pollutant, then which of the following is/are true about pollutant and its effects.
 - (A) SO_2 and NO_2 are major contributors to acid rain in the form of H_2SO_4 and HNO_3
 - (B) Chemical CF_2Cl_2 is common component of photochemical smog.
 - (C) Bacteria, Fungi, Moulds , Algae are example of viable particulants.
 - (D) CO_2 , CH_4 , O_3 , water vapour in the atomsphere causes global warming.
 2. Hydrogen is first element in periodic table , it may exist in the form of hydride, dihydrogen and other forms. Then which of the following statement(s) is/are correct regarding compounds of hydrogen element.
 - (A) Dihydrogen commercially produced by electrolysis of acidified water using platinum electrodes
 - (B) Ionic hydrides are good reducing agents
 - (C) In D_2O total number of electrons are greater than H_2O
 - (D) Fe_2O_3 and Cr_2O_3 are used as catalyst in Bosch process

3. (I) $\text{K}_3[\text{Fe}(\text{CN})_6]$ and (II) $\text{K}_4[\text{Fe}(\text{CN})_6]$ reagents are used in qualitative analysis of some of the basic radicals.
Then select the **INCORRECT** option(s).
- (A) Both the compounds follow Sidgwick E.A.N. rule
(B) Compound-I is diamagnetic where as Compound-II is paramagnetic in nature.
(C) Both compounds have d^2sp^3 hybridization with respect of central metal ion.
(D) Compound-I is more stable than Compound-II
4. (I) $[\text{Mn}(\text{CO})_6]^+$, (II) $[\text{Cr}(\text{CO})_6]$, (III) $[\text{V}(\text{CO})_6]^-$ are examples of organometallic compounds.
Select **CORRECT** option(s) regarding above given compounds.
- (A) All are isoelectronic carbonyl compounds
(B) In Compound-III carbon-oxygen bond length is maximum
(C) $[\text{Mn}(\text{CO})_6]$ act as reducing agent as compared to $[\text{Mn}(\text{CO})_6]^+$
(D) Metal-Carbon bond has least π -character in (III) among the Metal-Carbon bond of I, II and III
5. To get pure metal from impure metals, there are many refining processes according to nature of metal.
Select **CORRECT** option(s) regarding refining of metals.
- (A) Poling and electrolytic refining process can be used for refining of impure Cu.
(B) Poling, Liquation, Electrolytic refining is connected with refining of Sn metal
(C) Zone refining is based upon fractional crystallization, as the impurity prefers to stay in the melt. And it is used for refining of Si, Ge elements.
(D) Basemeerization process is used for refining of pig iron to form cast iron
6. Concentration of ore is necessary to remove unwanted particles, so which of the following option is/are **INCORRECT** in terms of concentration methods.
- (A) In leaching of red bauxite, hall process is used to remove impurity of Fe_2O_3 and NaOH solution is used as a leaching agent
(B) Magnetic separation is used to remove non-magnetic impurity of Wolframite from magnetic cassiterite
(C) Hydraulic washing is based on the different chemical properties of the ore and gangue particles
(D) Potassium or sodium ethyl Xanthate is mixed in frothing mixture as a froth stabilizers in froth floatation method

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
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Full Marks : +3 If ONLY the correct option is chosen.

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Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

Werner was able to explain the nature of bonding in complex, and he concluded that in complex the metal show two different valency

(1) Primary valency

(2) Secondary valency

7. For one of the isomer of $\text{PtCl}_2\text{I}_2 \cdot 4\text{NH}_3$, If we add excess AgNO_3 solution in 1 mole isomer of $\text{PtCl}_2\text{I}_2 \cdot 4\text{NH}_3$, then formation of silver halide precipitate take place, which is completely insoluble in NH_3 solution then formula of isomeric complex is.
- (A) $[\text{Pt}(\text{NH}_3)_4\text{I}_2]\text{Cl}_2$
- (B) $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{I}_2$
- (C) $[\text{Pt}(\text{NH}_3)_4\text{ClI}]\text{ClI}$
- (D) $[\text{Pt}(\text{NH}_3)_5\text{Cl}_2]\text{I}_2$
8. Which of the following statement is incorrect regarding $\text{Co}(\text{Br})(\text{Cl})(\text{I}) \cdot 4\text{NH}_3$.
- (A) 3 ionization isomers are possible
- (B) IUPAC name of the one of the ionization isomers is tetraamminebromidochloridocobalt(III) iodide
- (C) Primary valency of central metal ion is 3
- (D) Secondary valency is satisfied by only NH_3 .

Paragraph for Questions 9 and 10

In extraction of gold and silver, the respective metal is leached with dilute solution of NaCN or KCN in the presence of O_2 . Then the metal is obtained by replacement reaction.

9. Find out n-factor of Oxygen (O_2) and Gold (Au) respectively in reaction in which Gold complex is formed :-
- (A) 4, 1 (B) 4, 3
(C) 2, 1 (D) 1, 3
10. Which of the following is **NOT** related with metallurgy of Au or Ag.
- (A) Zn act as reducing agent and air act as oxidizing agent
(B) Both metal and carbon are sp hybridize in the complex of Ag and Au
(C) Complex of Zn, Ag and Au have tetrahedral, linear and linear geometry at metal respectively
(D) In the electrolytic refining of Silver, copper metal is obtain in anode mud but Zn metal is not

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

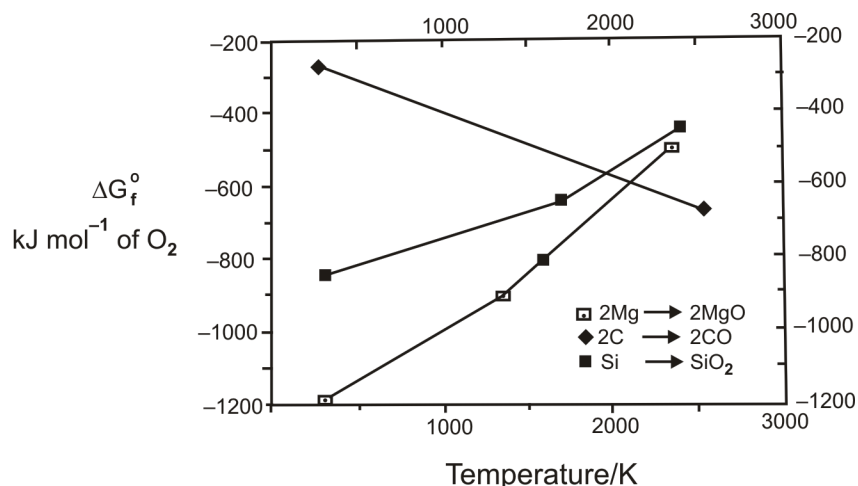
Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

The crystal field theory (CFT) is an electrostatic model, which considers the metal-ligand bond to be ionic, arising purely from electrostatic interaction between the metal ion and the ligand.

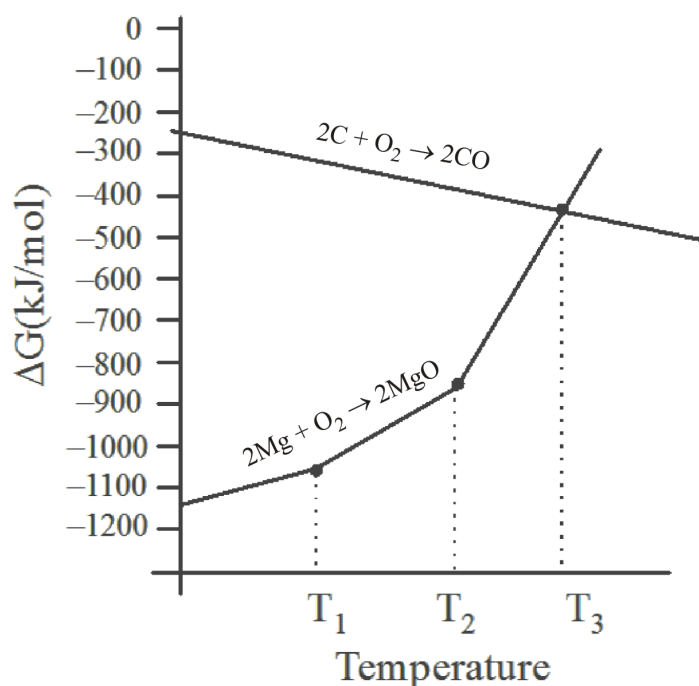
1. For complex $[\text{Co}(\text{CN})_6]^{-4}$
then find sum of $X + Y + Z$ (by considering C.F.T. only)
 X = Number of electron in t_{2g} orbital of central metal atom
 Y = Value of spin only magnetic moment
 Z = Number of electron in e_g orbital of central metal atom.
2. Experimental enthalpy of hydration of Cr^{2+} is $-460 \text{ k cal mol}^{-1}$. In the absence of crystal field splitting the value of enthalpy of hydration of Cr^{2+} is $-436 \text{ k cal mol}^{-1}$.
(Use $1 \text{ k cal mol}^{-1} = 350 \text{ cm}^{-1}$)
Find the value of $\left(\frac{\Delta}{10}\right)$ for $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ in cm^{-1} :

Paragraph for Questions 3 and 4



The free energy changes that occur when one gram molecule of a common reactant (in this case dioxygen) is used may be plotted graphically against temperature for a number of reactions of metals of their oxides. This graph is shown in figure and is called an Ellingham diagram (for oxides).

3. Select the number of **CORRECT** statement(s) about the plot is / are:



- (i) T_1 and T_2 are melting point & boiling point of Mg respectively.
 (ii) T_1 and T_2 are melting point & boiling point of MgO respectively.
 (iii) Reduction of MgO by coke is possible above T_3
 (iv) Mg can be extracted from gaseous products by rapid cooling.
 (v) After T_1 point $|\Delta G^\circ|$ decreases
 (vi) After T_2 point $|\Delta G^\circ|$ increases
4. As per given ellingham diagram (in paragraph) at 1500 K, if ΔG° for the oxidation of Si is -700 kJ/mole of O_2 and ΔG° for the reduction of MgO is $+850$ kJ/mole of O_2 .
 Find $|\Delta G^\circ|_{\text{redox}}$ for the reaction of
 $2\text{MgO} + \text{Si} \rightleftharpoons \text{SiO}_2 + 2\text{Mg}$ at 1500 K

Paragraph for Questions 5 and 6

Compounds which have the same molecular formula, but differ in their properties due to the difference in structure or orientation in 3D, are known as isomers.

5. A complex $[\text{CoBr}(\text{ONO})(\text{en})_2]\text{Cl} \cdot \text{H}_2\text{O}$, can show how many types of isomerism from given.
Linkage , Coordination , Ionization , Geometrical , Hydrate , Optical , Stereo isomerism
6. The octahedral complex $[\text{Co}(\text{en})_2(\text{NO}_2)_2]^+$ can exist in total number of isomeric form including structural & stereo isomers is/are :-

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

-
7. How many statements is/are **CORRECT** about H-atom and its compound(s).
- (i) Ortho $H_2 >$ Para H_2 : Stability at low temperature
 - (ii) $H_2 < D_2 < T_2$: Boiling point order
 - (iii) Atomic hydrogen $>$ Nascent hydrogen $>$ Molecular hydrogen : Reactivity order
 - (iv) Diborane is electron—precise covalent hydride
 - (v) Temporary hardness is due to presence of soluble salts of magnesium and calcium in the form of chlorides and sulphates
 - (vi) Equimolar mixture of CO and H_2 is known as water gas.
8. Pollution of water originates from Human activities. Select **CORRECT** points related to water pollution.
- (1) Clean water would have BOD value of less than 5 ppm where as highly polluted water would have a BOD value of 17 ppm or more.
 - (2) F^- concentration above 2 ppm cause brown mottling of teeth.
 - (3) Upper limit concentration of lead in drinking water is about 50 ppb
 - (4) Maximum limit of nitrate in drinking water is 50 ppm.
 - (5) Greater than 500 ppm of sulphate in drinking water causes Laxative effect
 - (6) 0.2 ppm of Fe and Al and 3 ppm of Cu metal is maximum prescribed concentration in drinking water.
9. In the following ores find sum of $(x + y + z)$ where
- x = number of ores in which two metal atoms exist in different oxidation state.
 y = number of ores in which two different metals are available.
 z = number of ores in which atleast one metal ion is diamagnetic in nature.
- Magnetite, Chromite, Limonite, Malachite, Iℓmenite, Chalcopyrite, Dolomite, Azurite, Carnalite, Cryolite

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. If $A = \begin{bmatrix} x + xy + xyz & 0 & 0 \\ 0 & y + yz + xyz & 0 \\ 0 & 0 & z + zx + xyz \end{bmatrix}$, $|A| \neq 0$ and $xyz = 1$ and sum of maximum

and minimum value of $\text{Tr}(A^{-1})$ (where $\text{Tr}(A)$ denotes sum of diagonal elements) is $\frac{p}{q}$, then which of following options is/are correct (p and q are coprime)

- (A) $p + q = 3$ (B) $p + q = 5$
(C) $p - q = 1$ (D) $3p - 2q = 4$
2. Let there are 6 hats of different colours and 6 shirts of same colours as that of hats. If the number of ways in which these can be put on by 5 men such that no man wear the shirt and the hat of the same colour is $K \times 6!$. Then the value of K is greater than
- (A) 108 (B) 218
(C) 258 (D) 309

3. Let $f(x)$ is a polynomial in x defined as $f(x) = (1 + x + x^2)(1 + x)^5(1 + x^2)^5$ then -

- (A) coefficient of x^{17} in $f(x)$ is 1
- (B) coefficient of x^{16} in $f(x)$ is 6
- (C) coefficient of x^{15} in $f(x)$ is 11
- (D) coefficient of x in $f(x)$ is 6

4. Let $B = [b_{ij}]_{n \times n}$ be a square matrix of order n such that $b_{ij} = \begin{cases} 0 & ; \quad i \neq j \\ \frac{4+i+j}{2} & ; \quad i = j \end{cases}$

(A) $\lim_{m \rightarrow \infty} \left(\frac{\sum_{n=1}^m \text{trace}(B)}{m^3} \right) = \frac{1}{6}$

(B) $\sum_{n=1}^{10} \text{trace}(B) = 330$

(C) $\sum_{n=1}^{10} \text{trace}(B) = 320$

(D) $\lim_{m \rightarrow \infty} \left(\frac{\sum_{n=1}^m \text{trace}(B)}{m^3} \right) = \frac{1}{3}$

5. $D = \begin{vmatrix} a & 1 & 1 \\ 1 & 1 & b \\ c & 1 & 1 \end{vmatrix} \times \begin{vmatrix} 1-b & bc-1 & 1-c \\ 0 & a-c & c-a \\ b-1 & 1-ab & a-1 \end{vmatrix}$. If $(a-c)(1-b) = 5$, then value of D will be -

(A) An even number

(B) An odd number

(C) A prime number

(D) A composite number

6. Let P is a skew symmetric matrix of order 3×3 with real entries. Given $I + P$ and $I - P$ (I is identity matrix of order 3×3) are non-singular matrices and $Q = (I + P)(I - P)^{-1}$ (Where $\det(Q) > 0$). Identify the correct options.

(A) $\det(2Q) = 8$

(B) $QQ^T = I$

(C) $\det(\text{adj}Q) = 1$

(D) $Q = Q^{-1}$

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

$$g(y) = \begin{vmatrix} p-y & q & r \\ r & p-y & q \\ q & r & p-y \end{vmatrix}, \text{ where } p, q, r \text{ are distinct positive numbers}$$

On the basis of above information answer the following :

7. Absolute term in $g(y)$ can be -
 - (A) 2
 - (B) -1
 - (C) 0
 - (D) All of the above
8. Coefficient of y in $g(y)$ will be -
 - (A) independent of only p
 - (B) independent of q as well as r
 - (C) independent of only r
 - (D) none of these

Paragraph for Questions 9 and 10

Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 3 \end{bmatrix}$ and U_1 , U_2 and U_3 are first, second and third columns of a 3×3 matrix U respectively.

If column matrices U_1 , U_2 and U_3 satisfy $AU_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $AU_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $AU_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$.

On the basis of above information, answer the following questions :

9. The value $|U|$ is

(A) $\frac{1}{6}$

(B) $-\frac{1}{3}$

(C) $\frac{1}{3}$

(D) $\frac{1}{2}$

10. U^{-1} is a

(A) Symmetric matrix

(B) Skew symmetric matrix

(C) Triangular matrix

(D) Scalar matrix

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Let $A6P + BQ4 + C45 = 887$, where 'XYZ' denotes three digits number.

On the basis of above information, answer the following questions :

1. If the number of ordered triplets (A, B, C) is λ then value of $\lambda/2$ is
2. If the number of solutions that product of A, B and C is an odd integer is μ then value of $\mu/5$ is

Paragraph for Questions 3 and 4

n straight lines are drawn in a plane such that no 2 of them are parallel and no 3 of them are concurrent.

Let T_n represents the number of parts in which the n lines divides the plane and let $Q_n = \prod_{r=1}^n \frac{T_{r+1} - 1}{T_r - 1}$.

3. T_{10} is equal to
4. $\sum_{r=1}^{10} \frac{6}{Q_r}$ is equal to

Paragraph for Questions 5 and 6

If $a_n = \sum_{r=0}^n \frac{1}{{}^nC_r}$ then $\sum_{r=0}^n \frac{r^2}{{}^nC_r} = P(n)a_{n+2} + Q(n)a_{n+1} + a_n + R(n)$

Where $P(n)$, $Q(n)$, $R(n)$ are polynomial functions of n then

5. $\frac{15}{10 - R(4)}$ is equal to
6. $P(5) + Q(5)$ is equal to

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. If A & B are two square matrices of order 3×3 such that $2A + 3BB^T = I$ and $B^{-1} = A^T$.
Let $\alpha = \det(A^{-1} - 3B^3 + BA)$ and $\beta = \text{Tr}(A^{-1} - 3B^3 - BA)$ then value of $\left(\frac{\alpha}{\beta}\right)$ is
8. The value of $\lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\sum_{t=0}^{r-1} \frac{1}{5^n} {}^nC_r {}^rC_t 3^t \right)$ is equal to
9. If a, b, c be the distinct real numbers and x, y, z be any three variables such that the system of linear equations
- $$(3z - 2y)a + (2x - z)b + (y - 3x)c = 0$$
- $$(y - 3x)a + (3z - 2y)b + (2x - z)c = 0$$
- $$(2x - z)a + (y - 3x)b + (3z - 2y)c = 0$$
- can be express as $AX = O$, A is 3×3 square matrix and $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ then the value of $\det(A) + \det(2A) + \det(3A)$ is :

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work

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